

Reading Poverty from AlphaEarth: Augmented Multidimensional Prediction and a Self-Supervised Poverty Representation

Preliminary results · work in progress · V1

Abstract

This paper builds a seven-country **H3 resolution-7** tabular spine for multidimensional poverty prediction by combining Google AlphaEarth embeddings with population, nightlights, vegetation, building, road, labelled with DHS-derived outcomes. The current analysis covers Ghana, Gambia, Kenya, Nigeria, Sierra Leone, Tanzania, and Zambia (*add more training cases in v2*). After spatial-block buffering, the effective labeled working set is **4,266 H3 resolution-7 cells** (4,219 for stunting), drawn from 5,186 total labeled cells.

The following questions are of interest: (i) **how well do AlphaEarth embeddings predict five DHS socioeconomic outcomes, and does augmenting them with additional socioeconomic information derived purely from Earth-observation sources further improve performance**, evaluated under strict spatial-block cross-validation and leave-one-country-out transfer; (ii) **how do different model families compare**—gradient boosting versus a graph neural network over H3 adjacency; and (iii) **can a self-supervised fused embedding, trained with no DHS information, reach comparable accuracy on the same prediction tasks**.

Results indicate that **AlphaEarth likely carries substantial socioeconomic signal**, strongest for wealth and women’s education, and that **explicit EO-derived socioeconomic features add substantial, transferable lift** for wealth, sanitation, women’s education, and mobile ownership; the lift is small for stunting. For wealth, the augmented model reaches $R^2 = 0.750$ under spatial cross-validation and 0.673 under leave-one-country-out transfer—results that are competitive with established EO-poverty benchmarks, though differences in spatial unit, country set, and evaluation scheme mean these should be read as indicative rather than as a like-for-like ranking. A single-task H3 resolution-7 GraphSAGE model is competitive with gradient boosting on the full feature matrix. The **self-supervised fused embedding—trained without any DHS labels**—seems to recover much of the supervised wealth and sanitation signal, falling only modestly short of the full supervised models. Leave-one-country-out transfer is strongest for wealth and sanitation, positive but weaker for women’s education and mobile ownership, and near zero for stunting.

1 Motivation And Prior Work

The broader project asks how far Earth observation and AI can go in estimating multidimensional poverty over space and time, and when a cheaper representation is good enough relative to raw satellite imagery. This version focuses on a narrower step: **can AlphaEarth be made more poverty-aware by adding interpretable, scalable geospatial features at the H3 resolution-7 level?** The contribution being tested is not that population density, roads, nightlights, or building counts are new. The contribution is the augmentation and evaluation design: AlphaEarth alone, tabular features alone, AlphaEarth plus tabular groups, graph models, and a DHS-free fused embedding.

The older satellite-poverty papers give a scale for what “good” has meant in related work, while the newer representation papers explain why a compact-embedding route is worth testing. Jean et al. established the satellite-to-poverty template by using imagery and machine learning to infer survey wealth. Yeh et al. scaled that logic with public imagery across ~20,000 African villages, reporting a held-out-country wealth R^2 of 0.67 on pooled cluster-level observations (or 0.70 when averaging per-country R^2). Pettersson et al. moved the problem toward temporal EO representation learning, using time series of satellite imagery across a larger DHS panel and reporting 72% of held-out-country and 75% of held-out-time variance explained for wealth. These papers motivate the benchmark question for this version: whether an H3 resolution-7 AlphaEarth-plus-tabular pipeline can approach the same wealth-prediction range while using precomputed embeddings and engineered socioeconomic layers rather than training a direct image sequence model.

The representation papers motivate a second question: whether the features can become a reusable embedding rather than just a supervised predictor. AlphaEarth Foundations motivates the 64-dimensional annual embedding as a compact general EO representation. Pettersson and Daoud use AlphaEarth with graph models for cluster-level poverty mapping across 37 DHS surveys and find that graph structure adds only a modest gain (best $R^2 = 0.569$ versus an image-only baseline of 0.546). Google PDFM and UrbanFusion are broader multimodal representation papers, not DHS poverty papers, but their methods are relevant because they combine spatial, human, environmental, and map-like modalities into reusable representations. MOSAIKS is an older precedent for shared satellite encodings with cheap downstream probes. In this paper, the frozen fused embedding is therefore read against the non-embedding models: XGBoost and GNN show what can be extracted with supervised access to the full feature matrix,

while the embedding probe asks what survives when DHS is not used to train the representation.

2 Data Construction

2.1 H3 Resolution-7 Spine, Countries, And Labels

The unit of analysis is an **H3 resolution-7 cell**. The GEE spine generates one row per H3 resolution-7 cell with country, H3 index, centroid coordinates, area, reference year, and polygon geometry. Admin-1 labels are assigned by centroid. This keeps the table simple and model-ready, but border cells should be audited if admin-level aggregation becomes part of the final claims.

The current analytic table contains **657,863 H3 resolution-7 cells** across seven countries. DHS labels are sparse relative to the full grid: 5,186 H3 resolution-7 cells have wealth, sanitation, women’s education, and mobile labels, and 5,132 have stunting labels. Ethiopia and Togo were present in upstream GEE feature generation but are not part of the current modeling set.

2.2 Source-Specific Feature Construction

The feature pipeline has two stages. The first builds the core satellite/composite features from Google Earth Engine: AlphaEarth, VIIRS, MODIS NDVI, and GHSL building height. The second processes WorldPop rasters, DHS labels, RWI audit fields, OSM roads, and Google Open Buildings morphology, then merges everything into a single H3 resolution-7 table.

AlphaEarth. AlphaEarth is treated as the core EO representation, not as a point sample. The extraction uses the annual `GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL` collection, top-coded to the latest available year when the DHS reference year is beyond source availability. The key implementation choice is polygon mean over each H3 resolution-7 cell rather than centroid pixel sampling. A centroid pixel would let a tiny point represent a multi-square-kilometer H3 resolution-7 cell; polygon means are slower but better aligned with the prediction unit. Chunked Drive exports and a 30 m reduction scale make this feasible.

- **Calculated features:** 64 H3 resolution-7 polygon-mean dimensions, `ae_A00`–`ae_A63`, plus AlphaEarth effective year.
- **Orthogonal features:** none. AlphaEarth is the base representation being augmented, not part of the orthogonal tabular block.

Nightlights. VIIRS is summarized as a 24-month signature ending in the reference year. Monthly images are masked by coverage, but raw radiance is not floored at zero before the mean and raw volatility are computed. This matters in dark

Table 1: Current analytic country set and DHS-labeled H3 resolution-7 cells.

Country	H3 res-7	Wealth	Sanitation	Women edu.	Stunting	Mobile
Ghana	60,433	602	602	602	600	602
Gambia	2,316	172	172	172	172	172
Kenya	106,744	1,490	1,490	1,490	1,489	1,490
Nigeria	190,814	1,350	1,350	1,350	1,313	1,350
Sierra Leone	17,349	464	464	464	454	464
Tanzania	158,287	596	596	596	592	596
Zambia	121,920	512	512	512	512	512
Total	657,863	5,186	5,186	5,186	5,132	5,186

rural areas, where small negative radiance values can carry calibration/background information. Log features use a tiny lower clamp only because logs require it.

- **Calculated features:** 24-month radiance mean, raw volatility, log slope, log volatility, valid-month count, and effective year.
- **Orthogonal features:** log slope and volatility. The slope is estimated over the 24-month log-radiance sequence; volatility captures instability rather than brightness level. Mean radiance is treated as an amount feature.

NDVI. NDVI was moved from Sentinel-2 to MODIS native 16-day products. Sentinel-2 had better spatial resolution, but for nine-country extraction it was slower, cloud-fragile, and memory-heavy in GEE. MODIS is coarser, but the denser annual sequence gives more stable phenology summaries.

- **Calculated features:** annual mean NDVI, robust seasonal amplitude, peak timing, valid-period counts, reliability flags, and effective year from 23 native 16-day periods.
- **Orthogonal features:** seasonal amplitude and peak timing. Amplitude is p90-p10 over the 16-day values rather than max-min, reducing sensitivity to single bad observations. Peak timing approximates when greenness is highest; NDVI mean is the amount-like vegetation feature.

WorldPop. WorldPop is processed locally from 100 m rasters. Local processing is used because raster no-data/sentinel values need explicit masking, and because rasterize-once plus bincount aggregation is much faster than polygon-by-polygon zonal statistics over many rasters and many H3 resolution-7 cells.

- **Calculated features:** population count, population density, youth dependency ratio, elderly dependency ratio, and sex ratio.
- **Orthogonal features:** youth dependency, elderly dependency, and sex ratio. These are derived from age-sex raster sums and are intended to capture demographic composition, not settlement amount. Population count/density are amount features.

GHS building height. GHS building height is a static 2018 layer in the selected source. It is therefore treated as a structural proxy rather than a year-matched DHS feature.

- **Calculated features:** mean, p90, and max built height within each H3 resolution-7 cell.
- **Orthogonal features:** height distribution. Verticality is not the same as footprint count; p90 and max height help distinguish dense low-rise settlement from denser vertical settlement.

Google Open Buildings. Google Open Buildings is filtered at confidence ≥ 0.70 . For Gambia, Ghana, Kenya, Sierra Leone, and Zambia, the direct Open Buildings S2 shard download path is used. Nigeria and Tanzania are processed locally from existing CSVs using the same morphology logic.

- **Calculated features:** building count, building density, footprint fraction, total area, mean/std building area, size Gini, nearest-neighbor spacing regularity, and orientation entropy.
- **Orthogonal features:** size Gini, spacing regularity, and orientation entropy. Size Gini summarizes inequality in footprint areas inside the H3 resolution-7 cell; spacing regularity is based on the variability of nearest-neighbor distances between building centroids; orientation entropy

summarizes whether footprints share a common direction or are more chaotic. Count, density, footprint fraction, and total area are amount features.

OSM roads. OSM road features are computed locally from GeoPackage extracts. Lengths and bearings are computed in a metric CRS, then segments are assigned to H3 resolution-7 cells by midpoint. This midpoint rule is fast and consistent, but it can under-attribute long roads that pass through a cell without their midpoint falling inside it.

- **Calculated features:** road density, road class maximum, segment counts, node counts, intersection counts, dead-end counts, dead-end ratio, degree shares, beta connectivity, grid-like diagnostic, and bearing entropy.
- **Orthogonal features:** dead-end ratio, degree shares, beta connectivity, and bearing entropy. These are topology/form features. Endpoint degree is computed from rounded segment endpoints, beta connectivity uses edges/nodes, and bearing entropy summarizes directional order. The hand-built grid-like diagnostic is dropped from model features so the model sees the raw topology components.

DHS and RWI. DHS outcomes are constructed from HR, IR, KR, and GE files, then aggregated to H3 resolution-7 cells. Outcomes are wealth, improved sanitation, women’s education, stunting, and mobile ownership. RWI is parsed and retained for audit completeness but is dropped before training, along with RWI/meta wealth columns, so the models do not use RWI as a shortcut.

Orthogonal feature logic. The orthogonal block is not literally independent of AlphaEarth. It is a deliberately imperfect diagnostic group: demographics, temporal variation, verticality, morphology, and topology are expected to be less reducible to a single visual intensity signal than population count, road length, nightlight mean, or building count. It remains incomplete. It does not yet capture institutions, service access, health systems, food security, disease ecology, or shocks. That incompleteness is useful to keep visible because it helps explain why stunting remains weak and why education/mobile transfer can be calibration-sensitive.

3 Training And Evaluation

3.1 Spatial Splits

Spatial leakage control is handled at the **block level**, not the H3 resolution-7 cell level. Each country is divided into one-degree longitude/latitude buckets, and each H3 resolution-7 cell is assigned to a country-specific spatial block from its grid indices plus a country prefix. This means a test H3 resolution-7 cell is not randomly surrounded by training cells from the same local area.

The fold assignment is greedy and label-aware. Blocks are sorted by labeled H3 resolution-7 load and total H3 resolution-7 load, then assigned to the fold with the lowest combination of label load, country imbalance penalty, and total cell load. Folds are accepted only if each active outcome clears the minimum label gate after buffering. The implemented final gate is 30 labels per fold.

Boundary buffering is applied after fold assignment. Approximate distance to the nearest one-degree block boundary is computed using latitude-adjusted longitude distances, and H3 resolution-7 cells within 5 km of a boundary are marked as buffer cells. The 5 km width is chosen to match the doc-

umented DHS displacement distribution: urban clusters are displaced by up to 2 km, and 99% of rural clusters by up to 5 km, with the remaining 1% up to 10 km. A 5 km buffer therefore covers all urban and 99% of rural displacement, while a 10 km buffer (reported for sensitivity in Table 2) is more conservative but discards nearly twice as many labels. Train/test masks then exclude buffered cells by default: training is non-buffer cells outside the held-out fold, and testing is non-buffer cells inside the held-out fold. Coordinates are also excluded from the feature matrix so the models cannot simply learn absolute geography.

For the GNN, the same fold logic matters twice. First, the held-out fold is spatially separated. Second, train and test subgraphs are built so message passing does not directly mix train and test H3 resolution-7 nodes across the split. This is why the graph result can be read as a spatial model rather than a leakage-prone smoothing exercise.

Table 2: Spatial split and buffer accounting. Counts are for the main DHS outcomes unless noted.

Setting	Used	Dropped	Drop %
10 km buffer, 1° blocks	3,466	1,720	33.2%
5 km buffer, stunting	4,219	913	17.8%
5 km buffer, 1° blocks	4,266	920	17.7%

3.2 Models And Why The Comparisons Matter

XGBoost is the main supervised tabular baseline, and it is the model the other approaches are measured against rather than a placeholder. Gradient-boosted trees remain the strongest general-purpose method on heterogeneous tabular data of this size: they handle mixed feature scales, nonlinearities, and high-order interactions without a graph, are stable on a few thousand labeled rows where neural models overfit, and require little tuning. This matters for the central comparison. A common expectation is that explicit spatial message passing should dominate a non-spatial tabular model; here, the spatial honesty already lives in the evaluation (block-level splits, displacement buffering, coordinate exclusion), so XGBoost is a spatially-evaluated but non-message-passing baseline. If a graph model cannot beat it under the same splits, the lift attributable to message passing specifically is small—which is the result we in fact observe. XGBoost is therefore not a weak foil but the bar the graph and embedding models must clear. Fold-safe preprocessing is fitted only on training data: missingness indicators are created from the training split, missing values are filled consistently, and robust scaling is learned on train then applied to test.

The GNN is a single-task GraphSAGE model over H3 resolution-7 adjacency. The original multitask GNN shared one output head across all DHS outcomes and was unstable, especially for mobile and stunting. The final model trains one outcome at a time, adds layer normalization, lowers hidden width, increases dropout, and uses an inner spatial validation split for early stopping.

The fused embedding is trained without DHS labels. It uses separate AlphaEarth and tabular encoders, fuses them into a latent Z of dimension 256, and trains with reconstruction, view-alignment, feature masking, and variance regularization. DHS evaluation happens only afterward through a frozen Ridge probe.

These comparisons are necessary for interpreting the embedding. XGBoost full is the supervised ceiling for the current feature table. GNN full asks whether H3 resolution-7 neighborhood aggregation adds signal beyond a strong tabular model. AlphaEarth-only shows the pure EO baseline. The frozen embedding asks whether the combined feature stack can be compressed into a reusable representation without DHS supervision. If the embedding is below XGBoost but above AlphaEarth on key outcomes, it is not a failed predictor; it is evidence about what information survives unsupervised fusion.

All leave-one-country-out results are reported as **pooled R^2** over concatenated held-out predictions rather than as a mean

of per-country R^2 . Pooling is deliberate: for the smallest countries (Gambia, Sierra Leone) and the weakest outcomes (mobile, stunting), single held-out countries contain too few labeled cells to yield a stable R^2 , and per-country values for those cells are large-magnitude and unreliable. Pooled R^2 is robust to this and is the only LOCO figure we treat as interpretable; per-country LOCO for small countries and weak outcomes should be read as diagnostic only.

4 Results And Discussion

Table 3: XGBoost pooled R^2 by feature set. Columns are ordered as standalone feature sets, combinations, then full stack.

Outcome	AE only	Tab only	AE+amount	AE+orth.	Full
<i>Panel A: spatial CV, pooled out-of-fold</i>					
Wealth	0.635	0.736	0.743	0.718	0.750
Sanitation	0.424	0.503	0.514	0.496	0.526
Women edu.	0.663	0.690	0.707	0.703	0.723
Mobile	0.468	0.501	0.504	0.498	0.516
Stunting	0.203	0.187	0.207	0.216	0.216
<i>Panel B: leave-one-country-out, pooled predictions</i>					
Wealth	0.505	0.657	0.682	0.643	0.673
Sanitation	0.281	0.385	0.415	0.387	0.410
Women edu.	0.171	0.329	0.275	0.311	0.306
Mobile	0.004	0.097	0.106	0.085	0.126
Stunting	-0.070	-0.042	-0.030	-0.008	-0.016

Interpretation: An interpretable outcome hierarchy emerges. Wealth is the strongest outcome, sanitation and women’s education form a middle-to-strong group, mobile ownership is locally predictable but weaker in transfer, and stunting remains near the floor. This ordering is substantively plausible: wealth is strongly expressed in settlement intensity, built form, roads, lighting, and population concentration; sanitation is partly infrastructural and urbanization-linked; women’s education is less directly visible but spatially correlated with the same long-run development gradients; mobile ownership is partly infrastructural but also depends on markets, prices, and network roll-out; and stunting is a biological and nutritional outcome that settlement structure should not be expected to capture. The hierarchy is not an artifact of one model—it holds across XGBoost, the GNN, and the frozen embedding—which is itself evidence that it reflects how visibly each outcome is written into the landscape rather than a quirk of estimation.

Wealth is strong preliminary evidence that augmented embeddings can map material poverty at low cost.

For wealth, the augmented pipeline is competitive with, and on the within-region metric exceeds, established EO-poverty benchmarks. The transfer-comparable figure is leave-one-country-out: our pooled LOCO wealth R^2 of 0.673 matches Yeh et al.’s pooled held-out-country 0.67 from public imagery, and approaches the 72% reported by Pettersson et al. from multi-temporal image sequences—both obtained here from precomputed annual embeddings plus tabular layers rather than direct image training. Under within-region spatial-block cross-validation our wealth R^2 rises to 0.750, above those headline numbers, though spatial CV and held-out-country measure different generalization regimes and should not be equated. Read as preliminary evidence, this points somewhere useful: a compact, frozen foundation embedding augmented with cheap, scalable geospatial features can recover material wealth at accuracy comparable to bespoke image models, at a fraction of the compute and storage. If this holds as the country set grows, it implies that fine-grained wealth maps—the input to geographic targeting of aid and services—can be produced and refreshed cheaply across data-scarce regions, which is the practical payoff the broader project is aiming at.

The tabular layers carry real, transferable signal that the embedding alone does not.

Adding the full tabular stack improves over AlphaEarth-only by +0.115 (wealth), +0.102 (sanitation), +0.060 (women’s education), +0.048 (mobile), and +0.013 (stunting) in spatial

CV, and by even more in transfer: +0.168, +0.129, +0.135, +0.122, and +0.054 respectively in LOCO. That the lift is *larger* out-of-country than within-region is the more important observation: the engineered socioeconomic layers do not merely fit local idiosyncrasy, they carry development signal that travels across borders better than the raw embedding does. The largest gains occur where population concentration, building amount, roads, and lighting are conceptually close to the target, and the smallest for stunting—consistent with the hierarchy above.

Built-environment “amount” and structural “context” features both add signal; neither is redundant with the embedding.

The feature split groups variables by how reducible they are to a single visual-intensity signal: amount features (population, building, light, and road intensity) versus orthogonal features (demographics, temporal variation, verticality, morphology, and topology). For wealth and sanitation, AE+amount edges out AE+orthogonal in both spatial CV and LOCO, so direct intensity variables still carry information AlphaEarth does not already encode. For women’s education and stunting the orthogonal block is competitive or ahead, suggesting that as outcomes become less visually expressed, non-intensity context contributes relatively more; demographics is the single most consistent contributor across outcomes. The clear reading is that AlphaEarth does not saturate even the amount-like built-environment signal—explicit socioeconomic layers add information on both axes, and the two groups are complementary rather than redundant with the embedding.

Table 4: Model comparison. Spatial values are pooled out-of-fold R^2 ; LOCO values are pooled across held-out countries.

Outcome	XGB full	GNN AE	GNN full	SSL Z
<i>Panel A: spatial CV</i>				
Wealth	0.750	0.649	0.751	0.706
Sanitation	0.526	0.422	0.513	0.467
Women edu.	0.723	0.659	0.706	0.546
Mobile	0.516	0.384	0.541	0.318
Stunting	0.216	0.216	0.204	0.160
<i>Panel B: leave-one-country-out</i>				
Outcome	XGB R^2	XGB ρ	SSL R^2	SSL ρ
Wealth	0.673	0.832	0.635	0.811
Sanitation	0.410	0.647	0.380	0.637
Women edu.	0.306	0.530	0.184	0.459
Mobile	0.126	0.478	-0.119	0.317
Stunting	-0.016	0.236	-0.095	0.056

Graph structure helps selectively, not universally.

GNN full is effectively tied with XGB full for wealth (0.751 vs. 0.750) and higher for mobile (0.541 vs. 0.516), while slightly lower for sanitation, women’s education, and stunting. H3 neighborhood aggregation appears to help when the outcome is spatially smoothed or network-like, but it does not automatically improve every target. Architecture matters: the original multitask GNN was unstable (mobile mean $R^2 = -0.283$, stunting 0.046), and the final single-task, layer-normalized, validation-stopped model raised mobile full to 0.541 and stunting full to 0.204—so respecting outcome heterogeneity, rather than forcing all DHS variables through one shared head, was necessary. That graph message passing only ties a non-spatial gradient-boosted model under identical spatial splits is itself informative: most of the spatial signal is already carried by the per-cell features and the honest evaluation, not by neighborhood aggregation.

Why a DHS-free augmented embedding, and what the result means.

The regressions above show that fusing AlphaEarth with tabular layers predicts poverty well—but they require DHS labels at training time, and DHS coverage is exactly what is scarce. The motivating question is therefore whether the *representation* itself can be made poverty-aware without consuming

labels, so that the expensive, sparse DHS signal is spent only on a light final probe rather than on learning the features. The frozen fused embedding tests this directly: AlphaEarth and tabular encoders are fused into a 256-dimensional latent Z trained only with reconstruction, view-alignment, masking, and variance objectives—never seeing a single DHS value—after which a frozen Ridge probe reads outcomes off Z .

The result is promising. On wealth the frozen embedding reaches $R^2 = 0.706$ in spatial CV and 0.635 in LOCO; the two most comparable points are our own supervised XGB full (0.750 / 0.673) and, externally, Pettersson and Daoud’s supervised AlphaEarth baselines (0.546–0.569 at cluster level). The label-free embedding thus sits only ~ 0.04 below a fully supervised gradient-boosted model that sees all 101 features and the DHS labels, and *above* the supervised image-only baselines reported in comparable work. In other words, most of the supervised-predictive signal for wealth survives compression into a representation that used no outcome labels at all; what is lost is fine predictive detail, not the broad socioeconomic gradient (Spearman $\rho = 0.811$ in LOCO). Sanitation behaves similarly (0.467 / 0.380 frozen vs. 0.526 / 0.410 supervised). The embedding is weaker for the less visually expressed outcomes, as expected.

What a good augmented embedding would unlock is the actual prize. A frozen, reusable Z that already encodes material-welfare structure means downstream poverty maps for a new survey, indicator, or country reduce to fitting a cheap linear probe on whatever few labels exist—no image handling, no re-training, no per-task feature engineering. That is the MOSAIKS/PDFM promise specialized to multidimensional poverty: a single task-agnostic representation, distributed once, queried many times. The present numbers do not yet establish that the embedding is production-ready—it trails the supervised ceiling, and it has been tested on seven countries—but they are preliminary evidence that the gap is small and that the route is worth pushing, which is what an early working result is meant to show.

Augmentation, not message passing, is the larger lever—consistent with prior graph-based work.

Pettersson and Daoud report cluster-level wealth R^2 of 0.546 for the AlphaEarth image-only baseline and 0.569 for their best graph variant, with graph structure adding only ~ 0.02 and their fuzzy-label variants performing worse than the plain baseline. Our AE-only wealth result is 0.635 spatial / 0.505 LOCO, and the augmented full model reaches 0.750 / 0.673. Spatial unit (their displaced cluster points with a 500 m buffer versus our H3 polygon means), country set (25 versus 7), and cross-validation scheme all differ, so these are indicative rather than head-to-head. But the two studies agree on the structural point—compact EO embeddings are already strong, and the extra increment comes more from what you add to them than from how you wire them together—and locate the larger increment in tabular augmentation rather than graph topology.

Wealth: the clearest success case.

AlphaEarth performs strongly, tabular-only performs better, and full XGB and full GNN both reach about $R^2 = 0.75$ in spatial CV with high LOCO. Tabular-only is close to full in LOCO (0.657 vs. 0.673), so the engineered socioeconomic variables already contain much of the transfer signal; AlphaEarth still adds value, but wealth is the outcome where visual structure and engineered layers are most aligned and mutually substitutable.

Sanitation: medium-strong and one of the better transfer outcomes.

XGB full reaches 0.526 spatial and 0.410 LOCO, with the frozen embedding close behind (0.467 / 0.380). The small supervised-to-frozen gap suggests sanitation is captured by broad settlement and infrastructure gradients rather than by highly outcome-specific signals. Population-accessibility

(gravity) features were already present and added negligibly over population density; travel-time-to-city and travel-time-to-health-service features were not included and are the cleaner accessibility ablation to test next.

Women’s education: spatially organized despite being invisible to imagery.

Women’s education is not directly visible, yet spatial CV is high (XGB full 0.723; AE-only already 0.663), indicating it is strongly organized through correlated socioeconomic gradients. Transfer is weaker but positive (0.306 LOCO), and rank information stays useful even where absolute calibration fails in particular countries—so future work should report ranking and calibration separately.

Mobile ownership: locally predictable, fragile in transfer.

Spatial CV is non-trivial (XGB full 0.516, GNN full 0.541): socioeconomic and infrastructure gradients carry signal even though mobile ownership is not a built-environment variable. LOCO is much weaker (0.126), reflecting country-specific telecom markets, pricing, handset diffusion, and survey timing. The spatial-CV result is meaningful; the transfer result is not yet evidence of robust generalization.

Stunting: the boundary case and a robust negative result.

Stunting is weak in every model family ($R^2 \approx 0.20$ for XGB, GNN, and the embedding alike) and near zero or negative in LOCO, and it does not move beyond ~ 0.22 even with the full tabular stack including population-accessibility. Among non-full combinations AE+orthogonal is strongest, consistent with the little available signal being carried by contextual structure rather than raw settlement intensity. The feature table likely lacks what should matter most for child nutrition—disease ecology, food security, maternal/child health-service access (travel-time-to-health, untested here), shocks, and survey-season controls—so stunting is best read as the negative case bounding how far settlement-structure features reach.

4.1 Toward Explainability

A central question for policy use is whether these models rely on meaningful built-environment signal or on country-level artifacts that happen to correlate with the outcome. Gain-based feature importances from the XGBoost full models give a first, coarse view of which inputs the supervised models actually use, but gain importance reflects only how often and how usefully a feature is split on—not its directional or marginal effect—and it is diluted across the 64 correlated AlphaEarth dimensions.

The intended next step, aligned with the broader project’s explainability objective, is to apply model-agnostic attribution (e.g. SHAP) and spatial residual analysis to test whether predictions rest on interpretable built-environment and demographic structure rather than on country identity. This matters most for the transfer (LOCO) setting, where reliance on country-specific artifacts would inflate within-sample performance while failing to generalize.

5 Current Limitations And Next Steps

Country coverage is still small. Seven countries are enough for a serious pipeline test, but not for broad Africa-wide claims. The next version should extend the country set once DHS/GPS files are consistent.

Temporal alignment is imperfect. AlphaEarth is available annually through 2024, VIIRS/NDVI can be year-matched more flexibly, but GHSL height is static at 2018 and some sources are top-coded.

The orthogonal/amount split is a first cut. The two groups contribute comparably, and the grouping is coarse: it does not yet separate institutions, service access, health systems, food security, disease ecology, or shocks, which likely matter most for the weaker outcomes. Refining the group-

ing and adding genuinely non-visual context is a natural next step.

Service-access features are only partially tested. Population-accessibility (gravity) features were included and were inert. Travel-time-to-city and travel-time-to-health-service features were not included in the reported runs and are the cleanest next accessibility ablation, particularly for sanitation and stunting.

LOCO calibration remains fragile. Pooled LOCO results are strongest for wealth and sanitation, but country-mean failures suggest that transfer is not solved. Ranking and calibration should be reported separately.

Next steps. Expand countries; run 0/5/10 km buffer sensitivity; add travel-time-to-city and travel-time-to-health-service features as their own ablation (distinct from the population-accessibility features already tested); split feature-family ablations beyond amount/orthogonal; improve the fused embedding with masked feature-group prediction, contrastive AE-tabular alignment, country-robust losses, and graph-aware objectives; adopt a displacement-aware (fuzzy) label scheme in the spirit of Pettersson and Daoud rather than a hard boundary buffer; compare against direct Sentinel-2, Landsat, and higher-resolution imagery where available; add explainability to test whether models rely on meaningful built-environment signals or country artifacts.

Provenance

All reported figures are derived from the same seven-country H3 resolution-7 feature table. Pooled metrics are computed over concatenated out-of-fold (spatial CV) or held-out-country (LOCO) predictions. Spatial-CV and LOCO results, the original multitask GNN comparison, and the final single-task GNN comparison are drawn from the corresponding prediction and metric records.

Linked Papers

1. *Combining satellite imagery and machine learning to predict poverty.* Neal Jean, Marshall Burke, Michael Xie, et al.
2. *Using publicly available satellite imagery and deep learning to understand economic well-being in Africa.* Christopher Yeh, Anthony Perez, Anne Driscoll, et al.
3. *Time Series of Satellite Imagery Improve Deep Learning Estimates of Neighborhood-Level Poverty in Africa.* Markus B. Pettersson, Mohammad Kakooei, et al., Adel Daoud.
4. *AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data.* Christopher F. Brown, Michal R. Kazmierski, Valerie J. Pasquarella, et al.
5. *Leveraging Compact Satellite Embeddings and Graph Neural Networks for Large-Scale Poverty Mapping.* Markus B. Pettersson, Adel Daoud.
6. *General Geospatial Inference with a Population Dynamics Foundation Model.* Mohit Agarwal, Mimi Sun, Chaitanya Kamath, et al.
7. *UrbanFusion: Stochastic Multimodal Fusion for Contrastive Learning of Robust Spatial Representations.* Dominik J. Muhlematter, Lin Che, Ye Hong, et al.
8. *A Generalizable and Accessible Approach to Machine Learning with Global Satellite Imagery.* Esther Rolf, Jonathan Proctor, Tamma Carleton, et al.